Data Collection & Product Report for 2025 Project: Camas Prairie Snow-On Survey

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Data Collection Summary:

Collection Dates, Flights:	1 flight on February 8, 2025 (DOY 039)
Aircraft, Equipment:	Piper PA-23-250 Aztec (N95TK), RIEGL VQ-580 II-S (H2229296)
Nominal Flight Parameters:	Flying Height: 600 m AGL, Speed: 120 kt, Swath Overlap: 50%
Nominal Equipment Parameters:	Pulse Rate: 600 kHz, Scan Rate: 160 Hz, Scan Angle: ± 37.5°
Collected Area:	128 km²

GNSS Reference Station Summary:

Station Name	Operating Agency	Coordinates (ITRF2014 epoch 2025.1055 / Ellipsoid)
CAMA	CRREL	43°17’59.29195” N, 115°18’01.84648” W, 1639.668 m
GSE2	NCALM	43°49’30.15062” N, 115°49’08.56837” W, 1182.131 m
P019	UNAVCO	43°18’00.70042” N, 115°18’41.99486” W, 1682.368 m
PILO	CRREL	43°57’34.23974” N, 115°41’10.21362” W, 2452.621 m

Data Processing Summary:

Scan Angle Cutoff:	N/A
Data Adjustments:	Line-by-line roll correction/elevation shift (using Find Match in TerraMatch), project elevation shift of +0.114 m to remove vertical bias (using kinematic ground control points)
Ground Classification:	1 iteration of aggressive ground determination (using TerraScan), manual classification of misclassified ground
Elevation Model Generation:	Bare-earth and first-surface models calculated from Kriging

Data Accuracy Summary:

Strip-to-Strip Average Magnitude:	0.032 m (using Measure Match in TerraMatch)
GCP Final Root Mean Square:	0.024 m (GCPs collected in project AOI)

Data Product Summary:

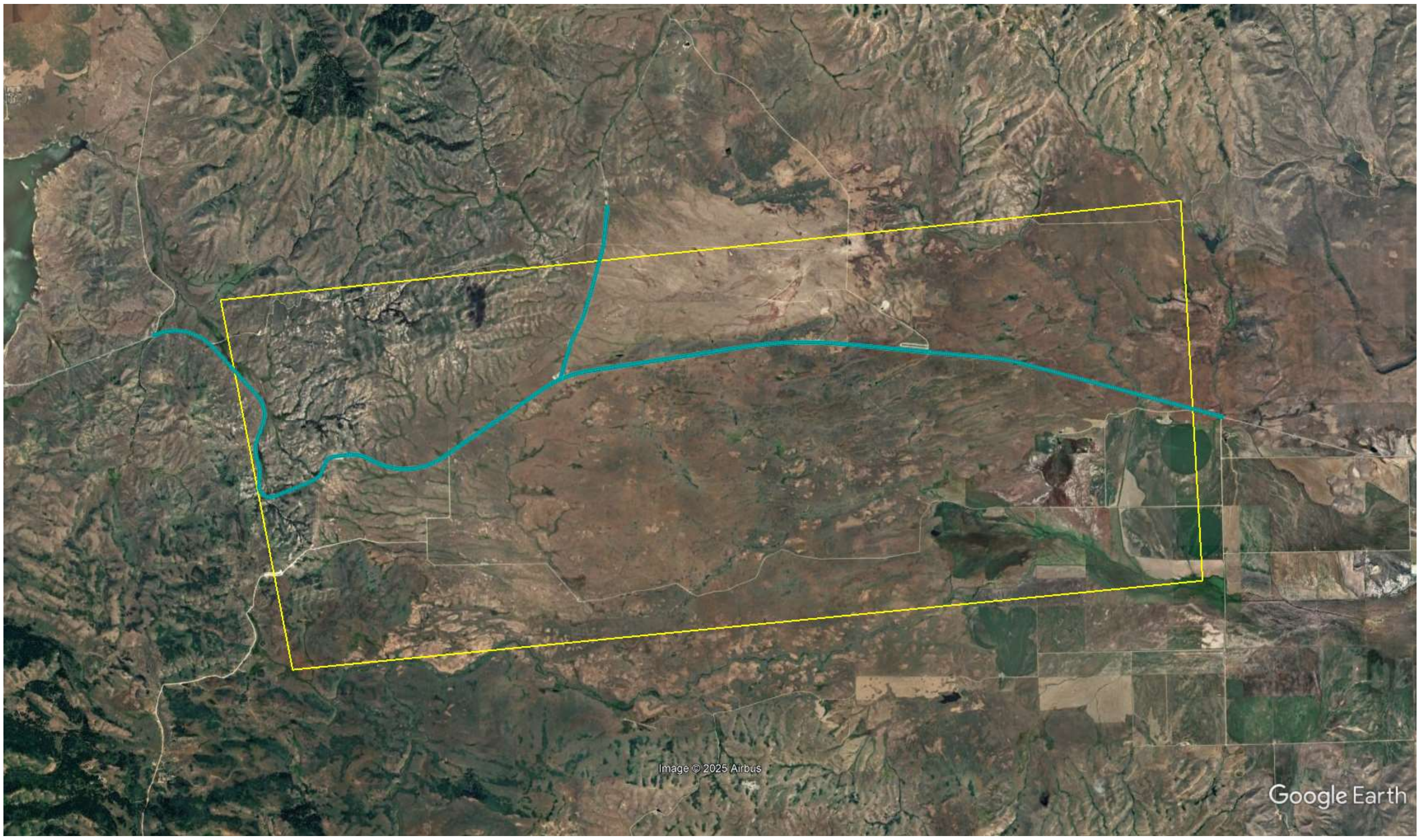
Horizontal / Vertical Datum:	WGS 84 (based on ITRF2014 epoch 2025.1055) / ellipsoid
Projection / Units:	UTM Zone 11N / meters (EPSG:32611)
Point Cloud Tiles:	1000-m × 1000-m tiles in LAS format (Version 1.4) with non-ground (1), ground (2), low noise (7), and high noise (18) returns
Bare-Earth Elevation Model:	GeoTIFF @ 1-m resolution from classified snow and ground
First-Surface Elevation Model:	GeoTIFF @ 1-m resolution with canopy and buildings included (from first-return of laser shots)

Area of Interest:

Location of survey polygon, aircraft trajectory, and GNSS reference stations

The requested survey area consisted of one polygon located southeast of Boise, ID. The polygon enclosed approximately 104 km^2 (40 mi^2). Two GNSS reference stations were located inside the area of interest.

Ground Control Points:



Location of ground control points in relation to area of interest

A total of 4,349 kinematic ground control points were collected inside the area of interest using a GNSS receiver and antenna mounted on a vehicle. GCPs were used to remove vertical bias in the final dataset.

APPENDIX 1: "Measure Match" Comparison to Surface Results

Line	Magnitude	Dz
1	0.0233	-0.0039
2	0.0326	0.0020
3	0.0318	-0.0026
4	0.0255	0.0024
5	0.0304	-0.0154
6	0.0246	0.0068
7	0.0326	0.0095
8	0.0303	-0.0025
9	0.0342	-0.0076
10	0.0288	0.0094
11	0.0426	0.0047
12	0.0380	-0.0152
13	0.0392	0.0057
14	0.0394	-0.0140
15	0.0356	0.0078
16	0.0300	0.0032
17	0.0301	0.0102

APPENDIX 2: "Output Control Report" Comparison to GCPs Results

Average Dz	0.000
Minimum Dz	-0.066
Maximum Dz	0.071
Average Magnitude	0.020
Root Mean Square	0.024
Standard Deviation	0.024